Day 172: Cross-Validation for Time Series and Grouped Data

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 25, Day 5 | Complete study guide

Watch: Cross-Validation for Time Series Forecasting (Egor Howell) | Your Random Train-Test Split May Be Training on the Future (Simply Machine Learning)

Script: `day172_cv_time_series_grouped.py` · every number in this guide is copied by program from `day172_run_output.txt`, the real output of that script.

Ordinary cross-validation deals rows into folds at random. When rows are neighbours in time, or belong to the same person, that lets a model score well by recognising what it has already seen. Today measures the damage against a known truth. On simulated series, shuffled folds rated a forecaster that copies nearby months at 0.780 when its real error was 2.104, and so chose the worst of three forecasters; forward-chaining folds chose the right one. On MovieLens, shuffled rows chose trees of depth 8 (0.9458), while folds that keep each user together and 300 users never seen both chose depth 1. The leak came from the users' own facts (age, sex, occupation), not mainly from the timestamp: with film facts alone it vanished.

0. Where today fits

DEFINITION — start here, no prior knowledge assumed

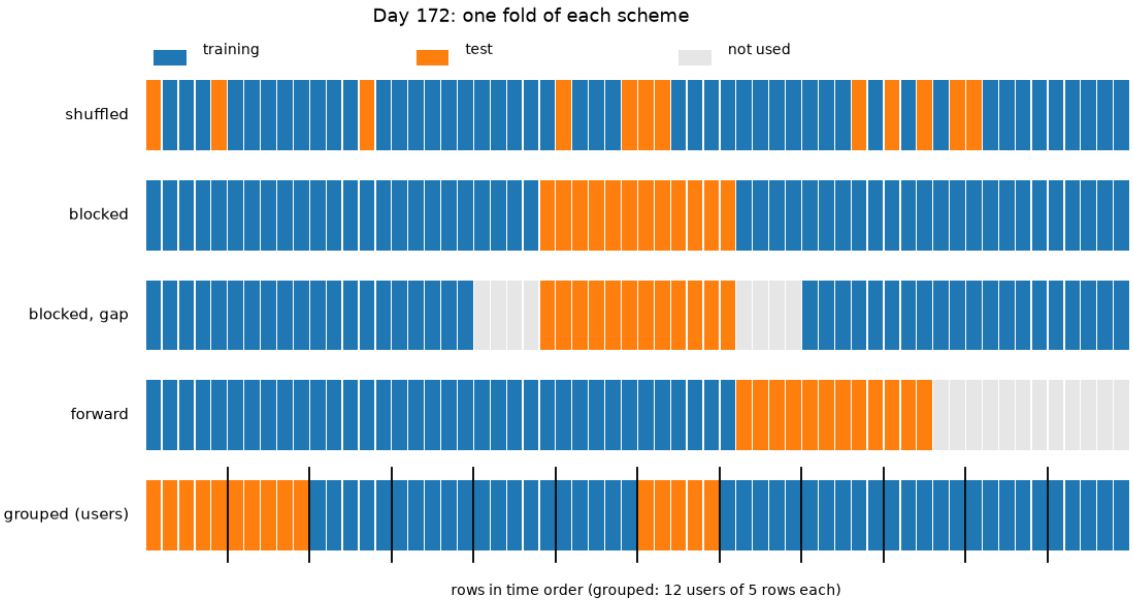
Cross-validation splits the rows into k folds; each fold in turn is the test set and the rest train. **Shuffled** folds deal rows at random. That is fair only if each row is independent of the others.

Time series: blocked folds cut the series into contiguous blocks; a **gap** leaves rows unused either side of the test block; **forward-chaining** folds train only on the past and test the next stretch, as a real forecast would. **Grouped data: group** folds keep every row of a person (a group) in the same fold, so test users are new users, as they will be in use.

REAL-WORLD SCENARIO

Testing a weather forecaster by hiding random days, then letting it look at the day before and the day after each one. It would look brilliant, and be useless for tomorrow. Or testing a voice assistant on new sentences from the same people it trained on, when it will be sold to strangers.

SYSTEM DIAGRAM



One fold of each scheme on 60 rows: training (blue), test (orange), unused (grey). The grouped strip shows 12 users of 5 rows each; black lines separate users.

Data: statsmodels' monthly CO2 (Mauna Loa, 1958 to 2001) and euro-area electrical-equipment orders (1995 to 2016), loaded with Week 22's code, and MovieLens 100K with Week 23's loaders. The final five years of each real series and 300 MovieLens users are held out from every fold and used only to check the estimates; nothing is chosen on them. The splitters are new; the boosted trees are Day 109's.

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
MATHEMATICS	The formula, when the section introduces one.
SPECIFICATION	The exact setup of the experiment.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. A known truth: simulated series

DEFINITION — start here, no prior knowledge assumed

50 series of 300 months, each a random walk with drift plus a yearly season plus noise. Because the recipe is known, the true error of a 12-month forecast made at the end of each series can be computed: simulate 400 possible next years and average the error.

Three forecasters predict a month 12 months after the last thing they may know. **trend**: a quadratic trend plus a level for each calendar month, fitted on the month number. **knn**: the mean of the 3 nearest months in the training rows. **lag**: least squares on the values 12 to 23 months earlier. Each is scored by the four schemes (5 folds; forward: five origins a year apart at the end of the series).

REAL-WORLD SCENARIO

Rehearsing a speech by reading every other sentence aloud with the script open: you will sound fluent, and it says nothing about the real talk.

MATHEMATICS

forward fold at origin o: train on rows with month ≤ o, test months o+1 ... o+12

true error = mean over 400 simulated futures and the 12 months of $|y_{future} - forecast|$

IMPLEMENTATION

```
def shuffled_folds(n, k, seed):
    f = np.random.RandomState(seed).permutation(n) % k
    return [(np.flatnonzero(f != j), np.flatnonzero(f == j)) for j in range(k)]
def blocked_folds(n, k, gap=0):
    b = np.arange(n) * k // n
    out = []
    for j in range(k):
        te = np.flatnonzero(b == j); lo, hi = te[0] - gap, te[-1] + gap
        tr = np.flatnonzero((b != j) & ((np.arange(n) < lo) | (np.arange(n) > hi)))
        out.append((tr, te))
    return out
def forward_folds(t, origins, horizon):
    t = np.asarray(t)
    return [(np.flatnonzero(t <= o), np.flatnonzero((t > o) & (t <= o + horizon))) for o in
        origins]
def group_folds(groups, k, seed):
    g = np.unique(groups); f = dict(zip(g, np.random.RandomState(seed).permutation(len(g)) % k))
    fold = np.array([f[x] for x in groups])
    return [(np.flatnonzero(fold != j), np.flatnonzero(fold == j)) for j in range(k)]
def cross_val(folds, fit_score):
    s = np.array([fit_score(tr, te) for tr, te in folds])
    return s.mean(), s
```

ACTUAL OUTPUT

```
  50 series of 300 months: random walk with drift + season + noise; truth from 400 simulated futures each
  forecaster           shuffled           blocked   blocked, gap 24           forward             truth
  trend                   1.013             1.783             2.407             1.453             1.244
  knn                     0.780             2.667             3.722             2.125             2.104
  lag                     0.972             1.080             1.162             1.058             0.989
  choice by shuffled        : knn
  choice by blocked         : lag
  choice by blocked, gap 24 : lag
  choice by forward         : lag
  choice by truth           : lag
```

What it means

Shuffled folds rewarded copying neighbours. knn scored 0.780 under shuffled folds, the best of the three, while its true error was 2.104, the worst. Shuffled folds would have chosen it.

For the lag model shuffled folds were nearly right (0.972 against 0.989). Its features are values from a year back, so a random test month's neighbours do not give its answer away. Bergmeir, Hyndman and Koo (2018) showed that ordinary k-fold is valid for purely autoregressive models when their errors are uncorrelated; this is that case.

Forward folds came closest and chose correctly: knn 2.125 against 2.104, lag 1.058. They were pessimistic for the trend model (1.453 against 1.244); a likely reason, not tested here, is that forward folds fit the quadratic trend on 240 to 288 months and extrapolate it, while the truth's model used all 300. Blocked folds with a gap were the most pessimistic: they train on less and test far from what they trained on.

WHY CHOOSE THIS?

Forward folds ask exactly the question a forecast answers, and here they chose the right model.

WHY NOT THIS? (TRADEOFFS)

They use fewer rows for training and fewer test points, so each estimate is noisier and a little pessimistic.

2. Two real series, with the final five years held out

DEFINITION — start here, no prior knowledge assumed

The same three forecasters and four schemes on CO2 and electrical-equipment orders, using months up to five years before the end. The held-out years are then forecast for real from five origins a year apart: one check of which estimate was honest. It is a single period, not a truth.

REAL-WORLD SCENARIO

Back-testing a trading rule on the last ten years, then watching it for five more before trusting it.

ACTUAL OUTPUT

```
CO2: 526 months, 1958-03 to 2001-12; folds use months up to 1996-12; held out 1997-01 to 2001-12
  trend  shuffled 0.613, blocked 1.293, blocked, gap 24 1.529, forward 1.709; held-out years 0.888
  knn    shuffled 0.919, blocked 3.693, blocked, gap 24 6.071, forward 3.125; held-out years 3.111
  lag    shuffled 0.466, blocked 0.551, blocked, gap 24 0.538, forward 0.647; held-out years 0.742
  choice by shuffled : lag
  choice by forward  : lag
  choice by held_out : lag
```

ACTUAL OUTPUT

```
electrical equipment: 257 months, 1995-01 to 2016-05; folds use months up to 2011-05; held out 2011-06 to 2016-05
  trend  shuffled 7.513, blocked 9.171, blocked, gap 24 11.415, forward 12.464; held-out years 8.835
  knn    shuffled 9.788, blocked 11.590, blocked, gap 24 18.496, forward 10.792; held-out years 7.098
  lag    shuffled 7.633, blocked 11.529, blocked, gap 24 10.633, forward 10.187; held-out years 2.907
  choice by shuffled : trend
  choice by forward  : lag
  choice by held_out : lag
```

What it means

CO2: every scheme chose the lag model, and the held-out years agreed (0.742). knn's shuffled estimate (0.919) was a third of its held-out error (3.111).

Electrical equipment: shuffled folds chose the trend model (7.513); forward folds and the held-out years chose the lag model. But the forward estimate for the lag model (10.187) was far above its held-out error (2.907); the next section shows why.

WHY CHOOSE THIS?

On real data the forward folds made the same choice as five unseen years.

WHY NOT THIS? (TRADEOFFS)

One series and one period: two real series cannot prove a rule, which is why section 1 exists.

3. Forward folds estimate a period

DEFINITION — start here, no prior knowledge assumed

The lag model's error from each forward origin, inside the folds and in the held-out years.

REAL-WORLD SCENARIO

A driving test taken in a snowstorm says something true, but about snowstorms.

ACTUAL OUTPUT

```
CO2: 1992-01 1.11, 1993-01 1.20, 1994-01 0.31, 1995-01 0.36, 1996-01 0.25; held-out origins 0.66, 1.17, 0.62, 0.81, 0.45
electrical equipment: 2006-06 8.23, 2007-06 4.52, 2008-06 14.16, 2009-06 16.92, 2010-06 7.11; held-out origins 3.40, 4.76,
    3.52, 1.98, 0.88
```

What it means

The electrical series' forward folds include the 2008 crash: the years starting 2008-06 14.16 and 2009-06 16.92 dominate the average, while the calm held-out years scored between 0.88 and 4.76. Forward cross-validation still ranked the models correctly, but its error level describes the years it tested. Report the per-fold errors, not only the mean.

4. Grouped data: ratings for new users

DEFINITION — start here, no prior knowledge assumed

MovieLens 100K. The task: predict the star rating a **new** user gives a film, from the film's facts (its mean rating and count in the training rows, its year) and the user's (age, sex, occupation) and the rating's timestamp. Model: Day 109's boosted trees (50 rounds, rate 0.2, at least 20 rows a leaf) at depths 1 to 8.

Folds use 400 users; 300 other users never enter a fold. Shuffled rows against folds grouped by user, each with 2 shuffles of 5 folds; the truth is a fit on all 400 users scored on the 300 new ones. Film means are always computed from the training rows only, so that is not a second leak.

REAL-WORLD SCENARIO

Marking a student's homework with questions from their own past homework: they recognise their own work.

MATHEMATICS

group folds: fold(row) = fold(user(row)); no user is in both the training and the test rows of any fold

IMPLEMENTATION

```
def ml_estimates(M, depth, use_time=True, use_user=True):
    dev = M["dev"]; fs = ml_scorer(M, depth, use_time, use_user)
    absf = lambda folds: [(dev[a], dev[b]) for a, b in folds]
    sh = np.mean([cross_val(absf(shuffled_folds(len(dev), K, s)), fs)[0] for s in
        range(CV_SEEDS)])
    gr = np.mean([cross_val(absf(group_folds(M["u"][dev], K, s)), fs)[0] for s in
        range(CV_SEEDS)])
    return sh, gr, fs(dev, M["new"])
```

ACTUAL OUTPUT

```
  43908 ratings from 400 users for the folds; 29961 ratings from 300 new users, never in a fold
  depth 1: shuffled rows 1.0230, grouped by user 1.0318, new users 1.0360
  depth 2: shuffled rows 1.0041, grouped by user 1.0370, new users 1.0390
  depth 4: shuffled rows 0.9750, grouped by user 1.0490, new users 1.0433
  depth 6: shuffled rows 0.9534, grouped by user 1.0657, new users 1.0498
  depth 8: shuffled rows 0.9458, grouped by user 1.0729, new users 1.0678
  choice by shuffled rows  : depth 8
  choice by grouped by user: depth 1
  choice by new users      : depth 1
```

What it means

Shuffled rows chose the wrong model. They said deeper was always better, down to 0.9458 at depth 8. New users said the opposite: depth 8 was their worst (1.0678) and depth 1 their best (1.0360).

Grouped folds told the truth. They chose depth 1 too, and stayed within about 0.016 of the new users' error at every depth (depth 1: 1.0318 against 1.0360).

A deep tree can single out one user from their facts and learn that user's personal average; with shuffled rows, that user's other ratings are in training, so it pays off in the score and never in use.

WHY CHOOSE THIS?

Group folds make every test user a new user, which is the question being asked.

WHY NOT THIS? (TRADEOFFS)

If the real task is predicting for known users (their next rating), shuffled or time-ordered folds are the right ones.

5. What carried the leak?

DEFINITION — start here, no prior knowledge assumed

The timestamp looked suspicious: sorted by time, consecutive ratings are nearly always the same user. The same estimates were repeated without it, and then with film facts only (no age, sex or occupation either).

REAL-WORLD SCENARIO

Finding which clue gave away the answer by covering them up one at a time.

ACTUAL OUTPUT

```
  depth 2, no timestamp   : shuffled rows 1.0098, grouped by user 1.0399, new users 1.0451; grouped minus shuffled +0.0301
  depth 2, film facts only: shuffled rows 1.0368, grouped by user 1.0355, new users 1.0365; grouped minus shuffled -0.0014
  depth 6, no timestamp   : shuffled rows 0.9719, grouped by user 1.0673, new users 1.0618; grouped minus shuffled +0.0954
  depth 6, film facts only: shuffled rows 1.0420, grouped by user 1.0417, new users 1.0382; grouped minus shuffled -0.0003
  ratings sorted by time: the next rating is by the same user 0.934 of the time
```

What it means

The timestamp was not the main carrier. Sorted by time, the next rating was by the same user 0.934 of the time, yet dropping it shrank the leak at depth 6 only from 0.112 (section 4) to +0.0954.

The users' own facts carried it. With film facts only, grouped minus shuffled was -0.0014 and -0.0003: no leak at all. A likely reason, not tested here, is that age, sex and occupation together pick out few enough users among 400 that a deep tree can recognise many of them.

For new users the user facts were worth almost nothing: film facts alone at depth 2 scored 1.0365, against 1.0390 with everything. Shuffled folds credited them with a skill that does not transfer.

WHY CHOOSE THIS?

Removing features one at a time locates a leak, instead of guessing.

WHY NOT THIS? (TRADEOFFS)

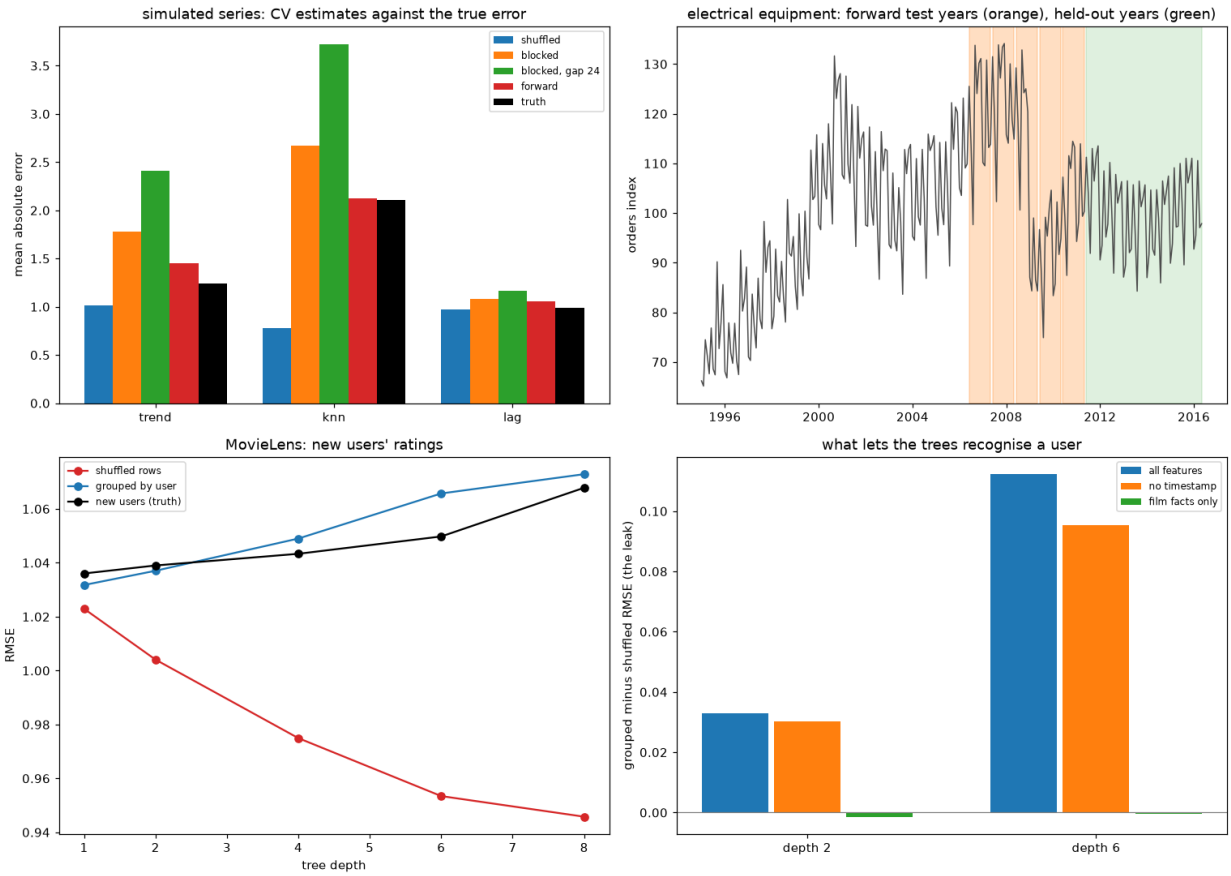
It costs one full cross-validation per feature set: here 21 model fits each.

Today's part for the week's build

`day172_exercise.py` holds `shuffled_folds`, `blocked_folds`, `forward_folds`, `group_folds` and `cross_val`, copied unchanged. Its last check is a small leak: 'copy the nearest training day' on a random walk, whose errors can be worked out by hand. Its hand checks:

ACTUAL OUTPUT

```
shuffled_folds(10, 5): test sizes [2, 2, 2, 2, 2], every row tested once: True (by hand [2, 2, 2,
    2, 2], True)
blocked_folds(10, 5), fold 3: test [4, 5], train [0, 1, 2, 3, 6, 7, 8, 9] (by hand [4, 5] and the
    other 8)
  with gap 1: train [0, 1, 2, 7, 8, 9] (by hand [0, 1, 2, 7, 8, 9]: rows 3 and 6 touch the test
      block)
forward_folds(0..9, origin 5, horizon 2): train [0, 1, 2, 3, 4, 5], test [6, 7] (by hand 0..5 and
    [6, 7])
group_folds on groups [0, 0, 1, 1, 1, 2, 3, 3, 4, 4], 2 folds: no group on both sides: True (by
    hand True)
cross_val with score = test size: mean 2.0, per fold [2, 2, 2, 2, 2] (by hand 2.0)
random walk, 'copy the nearest training day': shuffled folds 0.83, forward folds 3.04 (by hand:
    shuffled copies a neighbour usually 1 step away, so about E|N(0,1)| = 0.80, a little more when
    it is 2; forward copies a day 1 to 20 steps back, 0.80 x the mean of sqrt(1)..sqrt(20) = 2.46
    on average, from only 5 origins here)
```



Top left: simulated series, each scheme's estimate against the true error. Top right: the electrical series, with the forward test years and the held-out years. Bottom left: MovieLens by tree depth. Bottom right: the leak (grouped minus shuffled) with all features, without the timestamp, and with film facts only.

Interview preparation — technical questions

Q: How do you cross-validate a forecaster?

A: Forward chaining: train on the past, test the next horizon, from several origins; here it chose the right model where shuffled folds chose the worst.

Q: When is ordinary k-fold acceptable for a time series?

A: For purely autoregressive models with uncorrelated errors (Bergmeir et al. 2018); here the lag model's shuffled estimate was 0.972 against 0.989.

Q: How do you split data with many rows per user?

A: Group folds by user when the model will meet new users; shuffled rows chose depth 8, new users needed depth 1.

Q: How do you find which feature leaks?

A: Repeat the comparison dropping feature sets; here film facts alone removed the leak.

Q: Why report per-fold errors?

A: Forward folds estimate the periods they test; the 2008 crash folds were 14 to 17, calm years 1 to 5.

Interview preparation — theoretical questions

Q: Why does shuffling break cross-validation for dependent rows?

A: Test rows' neighbours or owners sit in training, so a model can score by recognising them.

Q: What does a gap between training and test protect against?

A: Features or targets that overlap in time; it costs training rows and makes the estimate pessimistic.

Q: Is forward cross-validation unbiased?

A: No: it trains on less data than the final model, and estimates the periods it tests.

Q: What question does a split answer?

A: The one its test rows pose: new users, the future, or both; choose the split to match how the model will be used.

Key takeaway and what's next

> **KEY TAKEAWAY**
> Cross-validation answers the question its folds ask. Shuffled rows ask how well a model recognises what it has nearly seen; forward folds ask how it forecasts; group folds ask how it does on new people. Here the wrong question chose the wrong model three times: on the simulated series, the electrical series and MovieLens.

- Simulated: knn's shuffled estimate 0.780 against a true 2.104; forward folds chose correctly.
- Electrical equipment: forward folds chose like the held-out years, but their level reflected the 2008 crash.
- MovieLens: shuffled rows chose depth 8 (0.9458); grouped folds and new users chose depth 1.
- Film facts only: the leak vanished (-0.0003 at depth 6).

Suggested watch order, if starting from zero

- 1. Cross-Validation for Time Series Forecasting (Egor Howell) — rolling-origin cross-validation for forecasting, with Python.
- 2. Your Random Train-Test Split May Be Training on the Future (Simply Machine Learning) — how a random split trains on the future, and the ways leakage gets in.

Day 173 preview

- **Experiment tracking.** Every run's settings, data version, seed, code version and scores written down automatically, so a result can be found, compared and trusted later.